NEET UG 2027: Master Biology Chapter Notes

Plant Physiology: Photosynthesis, Respiration & Growth Regulators

Prepared, Compiled & Authored by Satya Prakash · Strictly Calibrated to Rationalized NCERT & NMC Scope

1 Photosynthesis in Higher Plants

Chloroplast Architecture & Photosynthetic Pigments

Photosynthesis occurs in chloroplasts of green mesophyll cells:

- **Membrane System (Grana & Stroma Lamellae):** Responsible for trapping light energy and synthesizing ATP and NADPH (**Light Reactions / Photochemical Phase**).
- **Fluid Stroma:** Contains soluble enzymes for enzymatic reduction of CO_2 into carbohydrates (**Dark Reactions / Carbon Reactions / Biosynthetic Phase**).
- **Pigment Separation (Paper Chromatography):**
  - **Chlorophyll *a*:** Bright / blue-green; primary reaction center pigment.
  - **Chlorophyll *b*:** Yellow-green; accessory pigment.
  - **Xanthophylls:** Yellow; accessory pigment.
  - **Carotenoids:** Yellow to yellow-orange; accessory pigments; protect chlorophyll *a* from **photo-oxidation**.
- **Absorption vs Action Spectrum:** Maximum absorption by chlorophyll *a* occurs in the **blue** ($400-450\,\text{nm}$) and **red** ($650-700\,\text{nm}$) regions, showing maximal rate of photosynthesis (first demonstrated by T.W. Engelmann using *Cladophora* and aerobic bacteria).

Photochemical Phase: Z-Scheme & Photophosphorylation

Light-harvesting complexes (LHC) comprise hundreds of pigment molecules bound to proteins:

Photosystem I (PS I): Reaction center chlorophyll *a* has absorption peak at **700** nm (P_{700}).

Photosystem II (PS II): Reaction center chlorophyll *a* has absorption peak at **680** nm (P_{680}).

Photolysis of Water: $2H_2O \xrightarrow[Mn^{2+},Cl^-]{OEC} 4H^+ + O_2 \uparrow + 4e^-$

(Associated with PS II on inner thylakoid membrane).

Comparison of Non-Cyclic vs Cyclic Photophosphorylation:

Parameter	Non-Cyclic Photophosphorylation (Z-Scheme)	Cyclic Photophosphorylation
Photosystems	Both **PS II and PS I** operate in series	**Only PS I** is active
Electron Pathway	Non-cyclic (electrons from water reduce $NADP^+$)	Cyclic (electrons cycle back to P_{700})
End Products	**ATP, NADPH and** O_2 evolved	**ATP alone** (no NADPH, no O_2 evolution)
Site in Chloroplast	Grana thylakoids (appressed membrane)	**Stroma lamellae** (lack PS II and NADP reductase)
Wavelength Needed	Normal sunlight ($\lambda \leq 680\,\text{nm}$)	Active when light of $\lambda > 680\,\text{nm}$ is available

Chemiosmotic Hypothesis (Peter Mitchell)

ATP synthesis in chloroplasts is coupled to the generation of a **proton gradient across the thylakoid membrane**. Protons accumulate in high concentration inside the **thylakoid lumen**, dropping lumenal pH:

1. **Splitting of water** on inner side of membrane releases H^+ into the lumen.
2. **Plastoquinone (PQ)** acts as a mobile hydrogen carrier; removes protons from stroma and dumps them into lumen.
3. **NADP reductase enzyme** on stroma side removes protons from stroma to reduce $NADP^+$ to NADPH+

H^+.

Proton Motive Force: Protons break the gradient by passing through the transmembrane channel of F_0 **of ATP Synthase** into stroma $\implies$ drives conformational changes in F_1 **headpiece** to catalyze $ADP + P_i \rightarrow ATP$.

Biosynthetic Phase: C_3 Calvin Cycle Energetics

Discovered by Melvin Calvin using radioactive ^{14}C in algal photosynthesis (*Chlorella*).

$$\textbf{Phase 1: Carboxylation:}\ \text{RuBP (5C)} + CO_2 \xrightarrow{\textbf{RuBisCO}} 2 \times \text{3-PGA (3C)}$$

$$\textbf{Phase 2: Reduction:}\ 2 \times \text{3-PGA} + 2\,ATP + 2\,NADPH \rightarrow 2\,\text{Triose-P} + 2\,ADP + 2\,NADP^+$$

$$\textbf{Phase 3: Regeneration:}\ \text{Triose-P} + 1\,ATP \longrightarrow \text{RuBP} + 1\,ADP + P_i$$

$$\textbf{Per Single } CO_2 \textbf{ Fixed:} \quad 3\,ATP + 2\,NADPH \tag{1}$$

$$\textbf{Per Glucose (6 } CO_2\textbf{):} \quad 6\,CO_2 + \textbf{18}\,ATP + \textbf{12}\,NADPH \longrightarrow C_6H_{12}O_6 + 18\,ADP + 12\,NADP^+ \tag{2}$$

Hatch-Slack Pathway (C_4 Plants) & Photorespiration (C_2 Cycle)

1. Kranz Anatomy of C_4 Leaves (Maize, Sugarcane, Sorghum):

- Bundle sheath cells surround vascular bundles in large layers; possess thick walls impermeable to gas exchange, no intercellular spaces, and large agranal chloroplasts with abundant **RuBisCO** (lack PEPcase).
- Mesophyll cells have granal chloroplasts containing **PEPcase** (lack RuBisCO).
- Primary CO_2 acceptor is **Phosphoenolpyruvate (PEP, 3C)**, forming **Oxaloacetic acid (OAA, 4C)**.
- Malic acid / Aspartic acid transported to bundle sheath cells $\rightarrow$ decarboxylated to release CO_2 and pyruvate (3C).
- **Advantage:** RuBisCO operates at saturated CO_2 concentration, completely **preventing photorespiration**!
- **Energetic Cost of C_4 Pathway:** Requires $30\,ATP + 12\,NADPH$ per glucose molecule (extra $12\,ATP$ to regenerate PEP).

2. Photorespiration (C_2 Cycle / Glycolate Pathway): Occurs under high O_2, low CO_2, and high temperature when RuBisCO acts as an **oxygenase**:

$$\text{RuBP (5C)} + O_2 \xrightarrow{\text{RuBisCO}} \text{Phosphoglycerate (3C)} + \textbf{Phosphoglycolate (2C)} \tag{3}$$

Operates sequentially in three organelles: **Chloroplast $\rightarrow$ Peroxisome $\rightarrow$ Mitochondria**. A wasteful process that yields **neither ATP nor sugar**, but consumes ATP and releases CO_2!

NEET Trap: Blackman's Law & Factors Affecting Photosynthesis

- **Blackman's Law of Limiting Factors (1905):** If a chemical process is affected by more than one factor, its rate is determined by the factor which is nearest to its minimal value.
- CO_2 **Saturation Levels:** C_4 plants saturate at $360\,\mu L/L$; C_3 plants saturate only beyond $450\,\mu L/L$. Because current atmospheric CO_2 is ≈ 400 ppm, CO_2 is the major limiting factor for C_3 plants (greenhouse crops like tomatoes and bell peppers show enhanced productivity in CO_2-enriched atmospheres).
- **Light Saturation:** Occurs at **10% of full sunlight**. Except for plants in shade or dense forests, light is rarely a limiting factor in nature.

2 Respiration in Plants

Glycolysis (EMP Pathway) & Link Reaction

Discovered by Gustav Embden, Otto Meyerhof, and J. Parnas. Occurs in the **cytoplasm** of all living cells; universal to both aerobic and anaerobic organisms.

Phase 1: Preparatory: $\text{Glucose} \xrightarrow{1\,\text{ATP}} \text{G-6-P} \rightarrow \text{F-6-P} \xrightarrow{1\,\text{ATP}} \text{F-1,6-BP}$ (Invests 2 ATP)

Phase 2: Cleavage: $\text{F-1,6-BP (6C)} \xrightarrow{\text{Aldolase}} \text{PGAL (3C)} \rightleftharpoons^{\text{TPI}} \text{DHAP (3C)}$

Phase 3: Payoff Phase (x2):

$$2\,\text{PGAL} + 2\,\text{NAD}^+ \rightarrow 2 \times 1,3\text{-BPGA} + \mathbf{2}\,\text{NADH} + \mathbf{2}\,\text{H}^+$$

$$2 \times 1,3\text{-BPGA} + 2\,\text{ADP} \rightarrow 2 \times \text{3-PGA} + \mathbf{2}\,\text{ATP (SLP)}$$

$$2\,\text{PEP} + 2\,\text{ADP} \xrightarrow{\text{PK}} 2\,\text{Pyruvate} + \mathbf{2}\,\text{ATP (SLP)}$$

Net Glycolysis Yield: $\mathbf{2}\,\text{Pyruvate} + \mathbf{2}\,\text{ATP} + \mathbf{2}\,\text{NADH} + \mathbf{2}\,\text{H}^+$ (No CO_2 or O_2) (4)

The Link Reaction (Oxidative Decarboxylation of Pyruvate): Occurs in the **mitochondrial matrix**, catalyzed by *pyruvate dehydrogenase* multienzyme complex (requires Mg^{2+}, TPP, Lipoamide, NAD^+, CoA):

$$2\,\text{Pyruvate (3C)} + 2\,\text{CoA} + 2\,\text{NAD}^+ \longrightarrow \mathbf{2}\,\text{Acetyl-CoA (2C)} + \mathbf{2}\,\text{CO}_2 \uparrow + \mathbf{2}\,\text{NADH} + \mathbf{2}\,\text{H}^+ \quad (5)$$

Krebs Cycle (TCA Cycle) & Electron Transport System (ETS)

1. Tricarboxylic Acid (TCA) Cycle (Hans Krebs, 1937): Occurs in the **mitochondrial matrix**:

Per Acetyl-CoA (2C): $\text{Acetyl-CoA} + \text{OAA (4C)} + \text{H}_2\text{O} \xrightarrow{\text{Citrate Synthase}} \text{Citrate (6C)} + \text{CoA}$

Decarboxylations: $\text{Isocitrate} \rightarrow \alpha\text{-ketoglutarate (5C)} + \text{CO}_2 + \text{NADH}$

$\alpha\text{-ketoglutarate} \rightarrow \text{Succinyl-CoA (4C)} + \text{CO}_2 + \text{NADH}$

Substrate-Level: $\text{Succinyl-CoA} + \text{GDP} + \text{P}_\text{i} \rightarrow \text{Succinate} + \mathbf{1}\,\text{GTP/ATP} + \text{CoA}$

Oxidations: $\text{Succinate} \rightarrow \text{Fumarate} + \mathbf{1}\,\text{FADH}_2$ (Succinate dehydrogenase in inner membrane)

$\text{Malate} + \text{NAD}^+ \rightarrow \text{OAA} + \mathbf{1}\,\text{NADH} + \text{H}^+$

Net Yield of TCA Cycle (2 Acetyl-CoA per Glucose): $\mathbf{4}\,\text{CO}_2 + \mathbf{6}\,\text{NADH} + \mathbf{2}\,\text{FADH}_2 + \mathbf{2}\,\text{ATP/GTP}$ (6)

2. Inner Mitochondrial Membrane ETS Complexes:

- **Complex I: NADH dehydrogenase** (contains FMN and Fe-S centers); transfers electrons from NADH to Ubiquinone (CoQ).
- **Complex II: Succinate dehydrogenase** (contains FAD and Fe-S centers); transfers electrons from succinate to Ubiquinone.
- **Complex III: Cytochrome bc_1 complex** (Cytochrome b, Fe-S, Cytochrome c_1).
- **Mobile Electron Carriers: Ubiquinone** (in lipid bilayer) and **Cytochrome** c (small soluble protein attached to outer surface of inner membrane).
- **Complex IV: Cytochrome** c **oxidase** (contains Cytochrome a, a_3 and two copper centers, Cu_A and Cu_B); reduces terminal electron acceptor O_2:

$$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2e^- \longrightarrow \text{H}_2\text{O} \quad \text{(Metabolic water)} \quad (7)$$

- **Complex V: ATP Synthase ($F_0 - F_1$ complex).** Pass of $2H^+$ through F_0 produces 1 ATP.

Total Yield per Glucose: **36** or **38** ATP (or $30 - 32$ ATP by modern P/O) (8)

Respiratory Quotient (RQ) Formulations

$$\text{RQ} = \frac{\text{Volume of CO}_2\text{ evolved}}{\text{Volume of O}_2\text{ consumed}} \quad (9)$$

Carbohydrates (Glucose): $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \implies \text{RQ} = \mathbf{1.0}$

Fats (Tripalmitin): $2C_{51}H_{98}O_6 + 145O_2 \rightarrow 102CO_2 + 98H_2O \implies \text{RQ} = \dfrac{\mathbf{102}}{\mathbf{145}} = \mathbf{0.7}$

Proteins: $\text{RQ} \approx \mathbf{0.9}$

Organic Acids (Malic Acid): $C_4H_6O_5 + 3O_2 \rightarrow 4CO_2 + 3H_2O \implies \text{RQ} = \dfrac{\mathbf{4}}{\mathbf{3}} = \mathbf{1.33} > \mathbf{1}$

Anaerobic Respiration: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \implies \text{RQ} = \dfrac{\mathbf{2}}{\mathbf{0}} = \infty$

3 Plant Growth Regulators (Phytohormones)

Five Classic Phytohormone Classes: Functions & Applications

Hormone Class	Discovery / Chemistry	Key Physiological Functions & Agricultural Applications
Auxins	Charles & Francis Darwin (*Phalaris* coleoptile); isolated by F.W. Went from *Avena* coleoptile tip. Natural: IAA, IBA; Synthetic: NAA, 2,4-D.	• Promotes **apical dominance** (growing apical bud inhibits lateral buds; removal / decapitation induces bushy hedge). • Initiates root formation in stem cuttings. • Promotes flowering in pineapples. Prevents premature fruit/leaf drop, promotes abscission of older leaves. • **2,4-D** is widely used as a selective weedicide / herbicide to eradicate dicotyledonous weeds without affecting monocot plants. • Induces parthenocarpy in tomatoes.
Gibberellins	E. Kurosawa discovered in *Gibberella fujikuroi* (foolish seedling / *bakanae* disease of rice). > 100 known (GA_1, GA_2, GA_3 most studied). Terpene nature.	• Causes elongation of internodes; increases length of grape stalks and elongates apples. • **Bolting** (internode elongation prior to flowering) in rosette plants like cabbage and beet. • Increases sugarcane stem length, boosting yield by up to 20 tonnes per acre. • Accelerates malting process in brewing industry.
Cytokinins	F. Skoog discovered kinetin from autoclaved herring sperm DNA; Letham isolated **Zeatin** from corn kernels. Aminopurine derivative.	• Promotes rapid cell division; high cytokinin-to-auxin ratio promotes shoot differentiation. • Overcomes apical dominance; promotes nutrient mobilization and delays leaf senescence (**Richmond-Lang effect**). • Promotes adventitious shoot formation and lateral shoot growth.
Ethylene	H.H. Cousins established volatile substance from ripened oranges ripened bananas. Only **gaseous hormone**.	• Accelerates fruit ripening accompanied by a sudden rise in respiration rate (**Respiratory Climacteric**). • Source compound: **Ethephon** (releases ethylene slowly). • Promotes senescence and abscission of plant organs (leaves, flowers). • Promotes root growth and root hair formation, increasing absorption surface area. • Breaks seed and bud dormancy (initiates germination in peanut seeds, sprouting of potato tubers). • Promotes female flowers in cucumbers to increase yield.
Abscisic Acid (ABA)	Discovered as Inhibitor-β, Abscission II, and Dormin. Sesquiterpene derivative; acts as general growth inhibitor.	• Known as the **Stress Hormone**: stimulates rapid closure of stomata during water stress / drought. • Induces and maintains **seed dormancy**, preventing vivipary and premature sprouting. • Acts as a direct antagonist to gibberellins (anti-GA).

Photoperiodism & Vernalization Highlights

- **Photoperiodism (Garner & Allard):** Response of plants to periods of day/night.
  - **Short-Day Plants (SDP):** Flower when day length is less than critical photoperiod (e.g., Tobacco, Soyabean, Chrysanthemum).
  - **Long-Day Plants (LDP):** Flower when day length exceeds critical photoperiod (e.g., Wheat, Radish, Spinach).
  - **Day-Neutral Plants (DNP):** No correlation between light exposure and flowering (e.g., Tomato, Sunflower, Pea).
  - **Site of Perception:** The site of perception of light/dark duration is the **leaves**. The hypothetical hormonal substance responsible for flowering is **Florigen** (migrates from leaves to shoot apices).
- **Vernalization:** Quantitative or qualitative dependence of flowering on low-temperature exposure ($1-6°C$); prevents precocious reproductive development late in the growing season (e.g., winter varieties of

wheat, barley, rye; biennials like sugarbeet, cabbage, carrot). Perceived by the **shoot apical meristem**.